INTRODUCTION

While duration is a linear approximation of the sensitivity of a bond's price to changes in yield, the true relationship between a bond's price and its yield-to-maturity is a curved (convex) line. We introduce convexity as a complementary risk measure to improve bond price change estimates based on modified duration alone to account for this non-linear relationship. The convexity adjustment becomes more important when considering larger moves in yield-to-maturity and longer-maturity bonds. These lessons will also show how to estimate duration and convexity for a portfolio of bonds, as well as highlight limitations due to underlying assumptions.

Most of the examples and exhibits used throughout the reading can be downloaded as a Microsoft Excel workbook. Each worksheet in the workbook is labeled with the corresponding example or exhibit number in the text.

Category
Yield-Based Bond Convexity and Portfolio Properties

Learning Module Overview

- Convexity is a complementary risk metric that measures the second-order (non-linear) effect of yield changes on price for an option-free fixed-rate bond. The convexity adjustment adds to the linear price estimate provided by modified duration.
- Convexity is always positive for an option-free fixed-rate bond, such that estimated price increases from a decline in yields are higher than duration alone would suggest and estimated price decreases from an increase in yields are lower than duration alone would suggest. Therefore, convexity is valuable to investors.
- Convexity has the same relationship with bond features as duration: a fixed-rate bond will have greater convexity the longer its time-to-maturity, the lower its coupon rate, and the lower its yield-to-maturity.
- Money convexity expresses convexity in terms of currency units or percent of par for a position in a bond because it is the product of a bond's annual convexity and its full price.
- Portfolio duration and convexity can be calculated (1) as the weighted average of time to receipt of the aggregate cash flows or (2) by using the weighted averages of the durations and convexities of the individual bonds that make up the portfolio.
- While the first method is theoretically correct, it is difficult to use in practice. The second method is commonly used by portfolio managers but implicitly assumes parallel shifts in the yield curve, which are rare.

Learning Module Self-Assessment

These initial questions are intended to help you gauge your current level of understanding of the learning module.

1. The annualized convexity of a four-year, 2.8% (semiannual) coupon bond priced at par at issuance on 1 August 2035 and maturing on 1 August 2039 is *closest* to:

A. 7.624.
B. 16.413.

Solution:

B is correct.

Four-Year, 2.8% Semiannual Bond at Issuance

Fixed Coupon (%)	2.80
Periods p.a.	2
Price (per 100 Par Value)	100
Yield-to-Maturity (%)	2.8000

(1)	(2)	(3)	(4)	(5)	(6) = (2) × (5)	(7) = Col. 2 × (Col. 2 + 1) × Col. 5 × (1 + YTM/2)$^{-2}$
Period	**Time to Receipt**	**Cash Flow**	**Present Value**	**Weight**	**Time to Receipt × Weight**	**Convexity of Cash Flows**
1	1.0	1.4	1.380671	0.01381	0.01381	0.02686
2	2.0	1.4	1.361608	0.01362	0.02723	0.07946
3	3.0	1.4	1.342809	0.01343	0.04028	0.15672
4	4.0	1.4	1.324269	0.01324	0.05297	0.25759
5	5.0	1.4	1.305985	0.01306	0.06530	0.38105
6	6.0	1.4	1.287954	0.01288	0.07728	0.52611
7	7.0	1.4	1.270171	0.01270	0.08891	0.69179
8	8.0	101.4	90.726533	0.90727	7.25812	63.53176
			100.0000	**1.00000**	**7.62391**	**65.65133**
Annualized Macaulay Duration and Convexity					**3.81195**	**16.41283**

Annualized convexity is 16.41283 and results from the sum of Column 7 (65.65133) divided by the square of the periods per year (two periods per year: $2^2 = 4$).

2. Identify which of the following statements is true. The convexity adjustment:

A. is always negative for an option-free fixed-rate bond.
B. accounts for the first-order effect of yield changes on a bond.
C. is added to the bond price change estimate provided by modified duration.

Hide Solution

Solution:

C is correct. The convexity adjustment is added to the price change estimate provided by modified duration for a given change in yield-to-maturity:

$$\%\Delta PV^{Full} \approx (-AnnModDur \times \Delta Yield) + \left[\frac{1}{2} \times AnnConvexity \times (\Delta Yield)^2\right].$$

A is incorrect because convexity is always positive for an option-free fixed-rate bond.

B is incorrect because convexity accounts for the second-order effect of yield changes on a bond's price and duration accounts for the first-order effect.

Questions 3 and 4 relate to the following information.

An investor holds a GBP50,000,000 position in a 10-year, 3.50% fixed-coupon bond (semiannual) trading at par value. Annualized modified duration is 8.376, and annualized convexity is 81.701. The investor expects interest rates to decline by 100 bps.

A. 7.57%.
B. 8.01%.
C. 8.78%.

Hide Solution

Solution:

C is correct. The expected price change is calculated as follows:

$$\%\Delta PV^{Full} \approx (-AnnModDur \times \Delta Yield) + \left[\frac{1}{2} \times AnnConvexity \times (\Delta Yield)^2\right].$$

$$\%\Delta PV^{Full} \approx (-8.376 \times -0.01) + \left[\frac{1}{2} \times 81.701 \times (-0.01)^2\right].$$

$\%\Delta PV^{Full} \approx 0.0878 = 8.78\%$

4. The addition of money convexity to the estimated change in the value of the bond position using money duration alone is *closest* to:

A. GBP204,252.
B. GBP408,503.
C. GBP4,188,220.

Hide Solution

Solution:

A is correct. This question is asking for the money convexity adjustment. First we calculate money convexity:

$MoneyCon = AnnConvexity \times PV^{Full}$.

$MoneyCon$ = 81.701 × GBP50,000,000.

$MoneyCon$ = GBP4,085,033,604.

Then we calculate the money convexity adjustment:

$$MoneyConv.Adjustment \approx \left[\frac{1}{2} \times MoneyConvexity \times (\Delta Yield)^2\right].$$

$$MoneyConv.Adjustment \approx \left[\frac{1}{2} \times \text{GBP}4,085,033,604 \times (-0.01)^2\right].$$

$MoneyConv.Adjustment \approx$ GBP204,252.

5. A EUR100 million bond portfolio contains the following bonds:

Bond	Maturity (yrs.)	YTM (%)	Market Value	Annualized Modified Duration	Annualized Convexity
A	3	2.80%	EUR40,000,000	2.858	9.752
B	10	3.50%	EUR60,000,000	8.376	81.701

The expected percentage price change of the bond portfolio given a 50 bp increase in yield-to-maturity is *closest* to:

A. 3.018%.
B. 3.085%.
C. 3.151%.

Hide Solution

Solution:

A is correct. The weights of Bond A and Bond B are 40% and 60%, respectively. As a result, the portfolio weighted-average duration and convexity measures are 6.169 and 52.921, respectively. The expected portfolio percentage price change given a 50 bp increase in YTM is calculated as follows:

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$$\%\Delta PV^{Full} \approx (-6.169 \times 0.005) + \left[\frac{1}{2} \times 52.921 \times (0.005)^2\right].$$

$$\%\Delta PV^{Full} \approx -3.018\%$$

6. Using weighted-average duration and convexity measures to estimate bond portfolio interest rate risk:

A. is difficult in practice.
B. is the theoretically correct approach.
C. assumes a parallel shift in the yield curve.

Hide Solution

Solution:

C is correct. Using the weighted-average duration and convexity measures to estimate bond portfolio interest rate risk is easy in practice, but it is not the theoretically correct approach (using the weighted average of time to receipt of the aggregate cash flows is theoretically correct). It does, however, implicitly assume a parallel shift in the yield curve.

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Discussion

What do you want to discuss... (required)

Hi,

Can someone confirm if we need to compute the Annualized Convexity in the same way that we see in the "Learning Module Self-Assessment" part above?
The Macc Dur computation is alright but the 7th Column seems a little bit confusing.

Thank you.

Tags: convexity

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Created 6 months ago by Sayyed Ali Khurshid Khatib

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